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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**First Semester M.Tech.(CSE) (CBCS) Degree Examinations**  
January 2023

**22CSE114 - DESIGN THINKING**

Duration: 3 Hours

Max. Marks: 100

**Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.**

| Unit – I   |                                                                                        | Marks | BT* | CO* | PO* |
|------------|----------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.         | a) What are skills required for design thinking practitioners? Discuss.                | 6     | L*2 | 1   | 1   |
|            | b) Construct a diagram for product development process and Explain the process.        | 8     | L2  | 1   | 1   |
|            | c) Interpret the various types of Customer Needs.                                      | 6     | L2  | 1   | 1   |
| 2.         | a) Summarize the steps in designing market strategy.                                   | 8     | L2  | 1   | 1   |
|            | b) Explain the need and process for market research.                                   | 6     | L2  | 2   | 1   |
|            | c) Interpret elements of human-centered design (HCD) Process and Benefits of HCD.      | 6     | L2  | 1   | 1   |
| 3.         | a) Construct Concept Development Process diagram and describe the same.                | 8     | L2  | 2   | 1   |
|            | b) Outline the elements of PDS with a neat diagram.                                    | 8     | L2  | 2   | 1   |
|            | c) List the Design Thinking process adopted by different companies.                    | 4     | L1  | 2   | 1   |
| Unit – II  |                                                                                        |       |     |     |     |
| 4.         | a) How brainstorming technique helps in problem solving? Justify.                      | 6     | L3  | 3   | 1   |
|            | b) Discuss the components of House of Quality with matrix diagram.                     | 8     | L2  | 3   | 1   |
|            | c) What are the misconceptions about creativity?                                       | 6     | L2  | 3   | 1   |
| 5.         | a) What is Bench-marking? Why it is required? When is not required?                    | 8     | L2  | 3   | 1   |
|            | b) Infer Product Development Mission Statement Example: Screwdriver Project.           | 6     | L2  | 4   | 1   |
|            | c) Justify the tips for promoting creativity in a team.                                | 6     | L3  | 4   | 1   |
| 6.         | a) Summarize the Creative Problem Solving Techniques with examples.                    | 8     | L2  | 4   | 1   |
|            | b) Explain the steps to Constructing the House of Quality (QFD).                       | 6     | L2  | 4   | 1   |
|            | c) Discuss the need for Problem decomposition techniques.                              | 6     | L2  | 4   |     |
| Unit – III |                                                                                        |       |     |     |     |
| 7.         | a) What is Design for the Environment ? When and Why is created?                       | 10    | L2  | 5   | 1   |
|            | b) What are ENVIRONMENTAL ISSUES to be considered for Design for the Environment?      | 10    | L2  | 5   | 1   |
| 8.         | a) What are Qualitative and Quantitative tools for measure Design for the Environment? | 10    | L2  | 5   | 1   |
|            | b) Explain Design for Environment Process in product life cycle with chart.            | 10    | L2  | 5   | 1   |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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